

Bharat Raksha AI

Full Product Details

AI-Powered Criminal Network Analysis & Investigation Intelligence Platform

Problem Statement: SIH26189

Ministry of Home Affairs — Blockchain & Cybersecurity

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This document describes the complete product: SIH-aligned themes, technology stack, end-to-end investigation flow, every feature/route, Autopilot mechanics, build phases, demo script, and evaluator talking points. AI outputs are investigative leads only — not findings of guilt.

1. SIH-Aligned Topics

Criminal network analysis: Neo4j POLE graph, PageRank, communities, bridge detection, cross-case links

Cybercrime / cybersecurity intel: UPI mule chains, SIM-swap timing, phishing domains, IP/device clusters, crypto hops

Blockchain / evidence integrity: SHA-256 hashing of evidence + ledger-style custody / tamper checks

AI / NLP for investigations: FastAPI OCR + NER, Copilot RAG Q&A, Autopilot pipelines, structured briefs

Investigation workflow: Cases, evidence vault, RBAC roles, alerts, audit logs, PDF reports

Women safety (extension): Stalking patterns, escalation, proximity risk, trafficking indicators

Justice / court readiness (extension): Charge-sheet outline assist, court pack manifest, evidence story cinema

Legal framing: AI outputs are investigative leads, not findings of guilt. Charge-sheet and court materials are assistive drafts for IO/PP review.

2. Technology Stack

Frontend: Next.js 15, TypeScript, Tailwind, Cytoscape.js

Backend: Next.js API routes, Prisma ORM

Primary DB: PostgreSQL (cases, entities, CDR, transactions, alerts)

Graph DB: Neo4j (criminal network relationships)

Object storage: MinIO (S3-compatible evidence files)

Queue / cache: Redis

AI microservice: FastAPI (Python) — OCR + NER (port 8001)

Auth: NextAuth.js with RBAC (Admin, Investigator, Senior Officer, Auditor)

Architecture: Investigator browser !' Next.js (UI + API) !' PostgreSQL · Neo4j · MinIO · Redis · FastAPI · Evidence hash ledger.

Typical local URLs: Web app <http://localhost:3000> or [:3001](http://localhost:3001) · AI service <http://localhost:8001>

Common seeded investigator login: investigator@bharatraksha.gov.in / Invest@Bharat2026!

3. End-to-End Investigation Flow

- 1. Login with role-based access
- 2. Create a case (crime type; optional playbook such as cyber-financial or stalking)
- 3. Upload evidence — FIR, CDR CSV, bank transactions, location data, etc.
- 4. Ingestion extracts entities (PERSON, PHONE, UPI, VEHICLE, EMAIL, DOMAIN, amounts...) and relationships
- 5. Autopilot (optional) runs playbook modules: CDR, financial, cyber, conflicts, silence, MO twin, brief, etc.
- 6. Explore Network / Financial / Communication / Geo / Cyber / Women Safety / Conflicts
- 7. Use Copilot or Voice Partner for questions and actions
- 8. Generate reports, court pack, and charge-sheet outline for briefing / disclosure hygiene

4. Complete Feature Directory

4.1 Investigate

/dashboard — Command Center: Live case/entity/alert metrics + AI Digest (what changed since last visit)
/cases — Cases: Create and manage investigation cases
/cases/[id] — Case workspace: Evidence, graph, CDR, financial, geo, cyber/women tabs, Autopilot overview
/evidence — Evidence Vault: Cross-case evidence browser
/entities/matches — Entity Matches: Review and approve/reject identity merges

4.2 Intelligence

/network — Network Graph: Interactive criminal network exploration
/intelligence — MO & Intelligence: Cross-case links, MO twins, brief, graph analytics, What-If Time Machine
/mo-twins — MO Twin Matcher: Modus-operandi fingerprint and similar-case ranking
/conflicts — Conflict Radar: FIR/notes vs CDR/bank/geo contradictions
/twins — Case Twin Compare: Side-by-side shared entity overlap between two cases
/silence — Silence Detector: Actors active before crime window who go quiet afterward
/mirror — Mirror Identity: Phone/UPI/vehicle/IP/email fusion scoreboard

4.3 Safety & Cyber

/cyber — Cyber Center: Mule scoring, OTP/SIM-swap windows, phishing, device/IP clusters, crypto hops
/women-safety — Women Safety: Stalking, harassment escalation, proximity corridors, trafficking indicators
/alerts — Alerts: Live investigative alerts; Night-Watch can speak critical alerts

4.4 AI Partner

/copilot — Copilot: Evidence-grounded Q&A (RAG) + action tools calling real intel APIs
/brief — AI Case Brief: Structured EN/HI brief with POIs, health score, conflicts, next actions
/voice — Voice Partner: Speech-to-text !' same agent tools; text fallback if mic denied
/clues — Clues & Hypotheses: Ranked clues and alternate theories with confidence + evidence links
/coach — Investigator Coach: Why this matters / next step coaching cards

4.5 Justice

/court — Court Pack: JSON manifest of reports, evidence hashes, alerts, custody summary
/cinema — Story Cinema: Playable chronological story beats from live evidence
/chargesheet — Charge-sheet Assist: Draft outline with statute hints — not a filed charge-sheet

4.6 Admin

/admin/users — Users: User management
/admin/audit — Audit: Access / action audit trail
/settings — Settings: Account / app settings
/features — All Features: Capability directory linking every tool
Also on many intel hubs: AI Pinboard (local device pins for entities/clues).

5. Core Case Workspace (Baseline SIH Product)

- Evidence upload & ingestion with hash integrity
- Entity extraction and relationship building
- Network / graph sync into Neo4j
- Communication intelligence (CDR) — frequent contacts, bursts, co-locations
- Financial intelligence — fan-in/out, circular flows, high-value / suspicious patterns
- Geo / timeline — towers, locations, chronological events
- Copilot Q&A grounded in case evidence
- Alerts when patterns fire
- PDF investigation reports
- Autopilot Overview — Dual Mode, Investigation Health, playbook modules, brief snippet

6. Autopilot / Playbook System

Dual Mode

- Autopilot: playbook modules run automatically after ingest / on demand; Sync / Re-analyze demoted to Advanced

- Manual Expert: classic investigator controls stay primary

Crime playbooks (examples)

- Cyber financial / UPI fraud
- Phishing / social engineering
- Stalking / women-safety related cases
- Trafficking / missing indicators
- Broader violent / property crime packs

Typical modules

cdr, financial, cyber, women_safety, geo, graph, conflicts, silence, mo_twin, brief, report

Investigation Health

Score 0–100 based on evidence completeness and module coverage, plus gap list (e.g. upload CDR, resolve conflicts).

7. How Major AI Features Work

Copilot RAG: Retrieves case evidence snippets and answers only from known facts

Copilot Agent: Detects action intents (brief / cyber / conflicts / clues...) and calls real libs/APIs

Silence Detector: Compares pre- vs post-incident CDR/txn activity

Mirror Identity: Clusters PERSON !" PHONE/UPI/VEHICLE/IP/EMAIL via relationships

Case Twins: Shared entity-value overlap across cases

MO Twins: Vector fingerprint (CDR/txn/entity mix) + cosine similarity

Conflict Radar: Narrative dates/places vs digital activity mismatches

Cyber Center: Rule/heuristic scoring over live financial + CDR + entities

Women Safety: Pattern detectors over CDR/geo/notes keywords

Story Cinema: Ordered beats from meta, alerts, bursts, txns, geo

Court Pack: Manifest download (hashes + metadata), not a sealed ZIP of PDFs

AI Digest: Buckets new alerts since last visit (conflicts / cyber / women / financial / CDR...)

Night-Watch: Optional TTS for critical new alerts while app is open

8. Build Phases (Expansion Pack)

Phase 0: Baseline protect: login, cases, ingest, graph, CDR, financial, geo, copilot, alerts, reports

Phase 1: Left sidebar AppShell + feature directory

Phase 2: Playbooks, Dual Mode, Autopilot pipeline, Investigation Health

Phase 3: Agent tools, Voice Partner, Clues, Hypotheses, Coach

Phase 4: Cyber Center + Women Safety suite

Phase 5: Silence, Mirror, Twins, Cinema, Conflicts UI, Charge-sheet, Court Pack, Pinboard, What-If, Night-Watch

Phase 6: MO Twin UI, Case Brief EN/Hi, Dashboard AI Digest, full regression

9. Suggested SIH Demo Script (5–8 minutes)

1. Login !" Dashboard !" show AI Digest and live stats
2. Open a case !" show uploaded FIR + CDR + bank evidence
3. Run Autopilot !" show Health score and completed stages
4. Open Network Graph and Financial anomalies
5. Open Cyber Center (mule / SIM-swap style leads)
6. Open Conflict Radar (FIR vs digital timeline)
7. Ask Copilot: Summarize this case / Brief do
8. Show Court Pack download + Charge-sheet Assist
9. Mention evidence hash / ledger for the blockchain theme

10. What to Emphasize for Evaluators

- Solves a real MHA investigation pain: multi-source digital evidence fusion
- Graph + AML-style financial + CDR analytics in one case OS
- Blockchain angle = evidence integrity / custody hashing
- Extra AI layers reduce manual busywork (Autopilot, Voice, Brief, Digest)
- Safety & Cyber modules address high-priority Indian crime categories
- Honest UX: leads + review required — no fake CCTNS/SOS integrations

